



TOWARDS THE REBUILDING OF SPINY LOBSTER STOCKS IN BRAZIL: A REVIEW

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ABSTRACT

In this study, we used the Monte Carlo model (CMSY) to estimate fisheries reference points based on annual lobster landings. During the period 2012-2018, fishing pressure on lobster stocks decreased and a tendency for recovery was observed, despite the low biomass levels of approximately 29%. In 2019, the stock size was capable of producing yields close to the maximum sustainable yield (5770 t), with a probability of 58.1%. The enforcement of management measures should be continued in order to eliminate sources of negative externalities and rebuild stocks. We also demonstrate that the closed season does in fact protect the main events of the lobster life cycle. Responsible and efficient management should focus on life cycle studies and reliable monitoring of fishing data.

Key words. — Monte Carlo method, spiny lobster, life cycle, stock assessment externalities, Brazil

RESUMO

O modelo de Monte Carlo (CMSY) foi utilizado para estimar os pontos de referência da pesca com base em desembarques anuais de lagosta. Embora se registrem os baixos níveis de biomassa de aproximadamente 29%, a pressão de pesca nos estoques de lagosta diminuiu e foi observada

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uma tendência de recuperação durante o período 2012-2018. Em 2019, o tamanho do estoque foi capaz de produzir rendimentos próximos ao rendimento máximo sustentável (5770t), com uma probabilidade de 58,1%. A aplicação das medidas da gestão deve continuar, com vistas a eliminar fontes de externalidades negativas e recuperar os estoques. Também é demonstrado que o defeso, de fato, protege os principais eventos do ciclo de vida da lagosta. A gestão responsável e eficiente deve se concentrar nos estudos do ciclo de vida e no monitoramento confiável dos dados da pesca.

Palavras-chave. — Método de Monte Carlo, lagosta espinhosa, ciclo de vida, avaliação de estoque, externalidades, Brasil

INTRODUCTION

Spiny lobsters (genus *Panulirus*) are the object of major commercial and recreational fisheries in the Wider Caribbean and along the coast of Brazil, using a variety of techniques and artisanal fishing gear (Briones-Fourzán & Lozano-Álvarez, 2000; Cruz & Phillips, 2000; Ehrhard, 2000; Fonteles-Filho, 2000; Hunt, 2000; Cruz & Borda, 2013). However, little information is available on selectivity, abundance, catchability, productivity, or catch-and-effort (Arreguín-Sánchez, 1996). The wide variety in fishing methods, boats and gear makes it difficult to study and monitor lobster fisheries and to standardize the unit of effort.

The Brazilian red spiny lobster, *Panulirus argus* (Latreille, 1804) was identified as *P. argus westonii* by Sarver et al. (1998) and as *P. meripurpuratus* by Giraldez & Smith (2016). In the Caribbean, the species was identified as *P. argus argus* (cf. Sarver et al., 1998). To avoid taxonomic confusion, in this paper we use the name *P. argus* for both Caribbean and Brazilian red spiny lobsters.

P. argus is an abundant and valuable fishing resource in the tropical western Atlantic. The green spiny lobster, *Panulirus laevis* (Latreille, 1817), though rare in the northwestern Atlantic, is the second-most important lobster species in Brazilian commercial fisheries, accounting for 29% of the total landings of spiny lobsters. Other species (*Panulirus echinatus* Smith, 1869, and four species of slipper lobster) are occasionally caught and sold on the domestic market (Ivo et al., 1996).

The management of Brazilian lobster fisheries is very challenging. For example, fishing zones have not been established, making it impossible to control catches by species and effort. Moreover, the lobster production, wholesale and marketing chain is complex, as shown by Santana et al. (2015), Cavalcante et al. (2011) and Ximenes & Vidal (2018), respectively.

Assessments of fishing resources based on fragmentary artisanal fishing data have historically been used in the mathematical modelling of commercial resources (in some cases violating model assumptions). This paradigm is no longer tenable. More reliable information is needed on the recruitment and relative abundance of

lobsters in the benthic ecosystem and on catches of spiny lobsters on the fishing grounds.

The methodology developed by Beddington & Kirkwood (2005) for the estimation of potential yield and stock using life-history parameters appears to be a useful approach. Another interesting approach is the prediction of recruitment and potential yield by monitoring different stages (puerulus, juvenile, adult) in the lobster's life cycle (Cruz et al., 2007b).

Recruitment, future catches and effort levels can be predicted using indices independent of fishing data. The first model was developed for *Panulirus cygnus* George, 1962 in the late 1980s in Western Australia (Phillips, 1986). Other models have been proposed that use large samples of pueruli, juveniles and environmental data to predict catches (Caputi et al., 1991; Caputi et al., 1995). In the Cuban Archipelago, models have long been used to assess and predict recruitment and catch based on lobster life cycle data collected with different artificial shelters (Cruz & Adriano, 2001; Cruz et al., 2001; Cruz, 2002).

In this review, we evaluate the depletion of Brazilian lobster stocks associated with overfishing, analyse the behaviour of fishermen and resource managers, and discuss stock rebuilding prospects. Our observations are interpreted in the light of experience from the Wider Caribbean.

DATABASE AND METHODS

Site description

Red and green lobster fisheries are concentrated in three regions along the Brazilian coast: the Great Amazon Reef System (GARS), the Northeast (NE) and the Southeast (SE). The management and exploitation of lobster resources are carried out exclusively by Brazilians (fig. 1).

GARS is a 1000 km long mesophotic reef ecosystem about 80 km off the Amazon River mouth, covering an area of 9500 km². The depth ranges from 30 m to 90-120 m at the shelf break (Moura et al., 2016). The large spiny lobster stocks in this region are exploited by commercial fishing vessels at a depth of 50-100 m (Cruz et al., 2014). The fishing grounds in NE are much shallower (3-60 m), but the shelf is wide, with extensive calcareous algae banks, possibly due to the absence of major rivers and, consequently, very small amounts of fluvial discharge (Coutinho & Morais, 1970). At a depth of approximately 60 m, the continental shelf drops off abruptly. In contrast, the shelf is narrow in SE, and the area accounts for the smallest landings of the three regions.

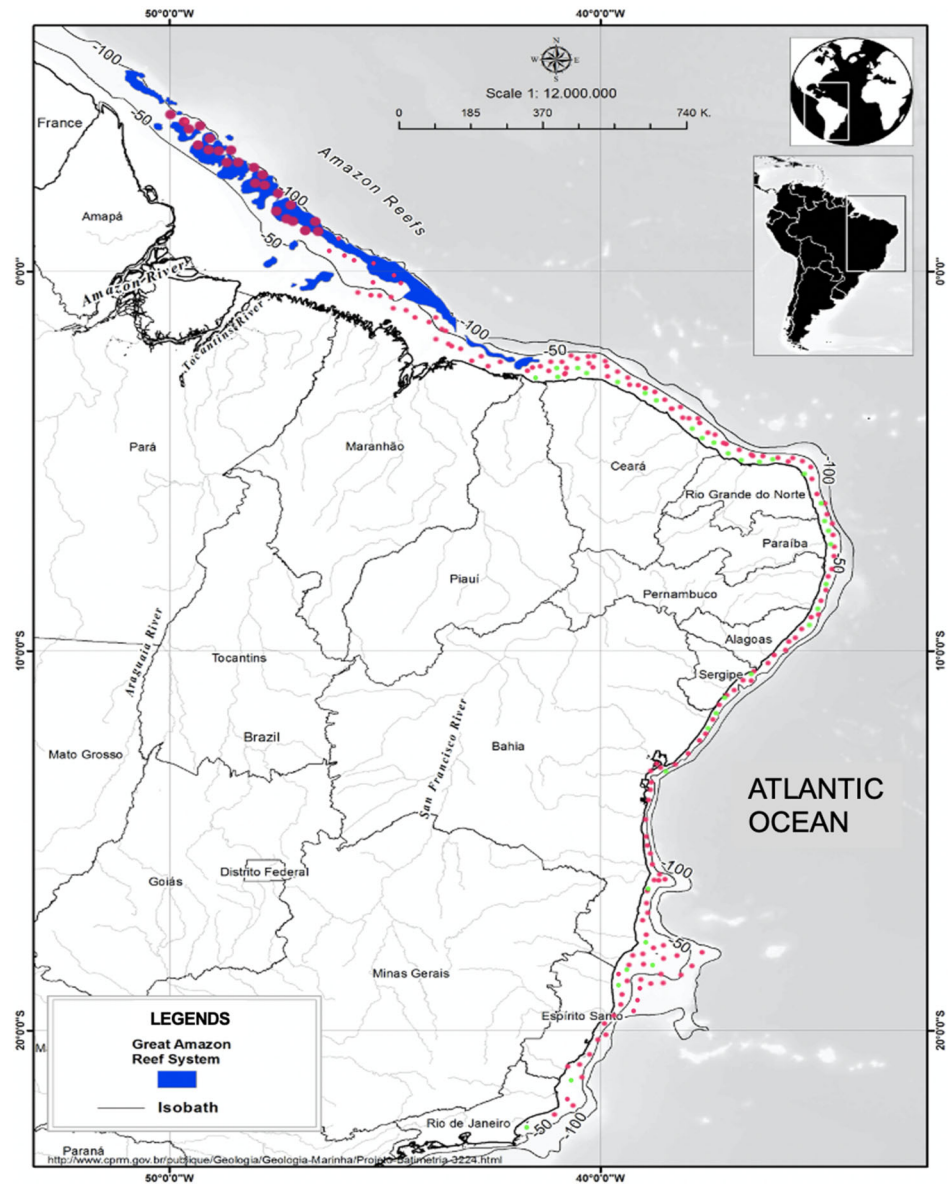


Fig. 1. Map of the continental shelf of Brazil showing fishing grounds (1-100 m) of the red spiny lobster (red dots), *Panulirus argus* (Latreille, 1804) and the green spiny lobster (green dots), *Panulirus laeviscauda* (Latreille, 1817).

Time series

The method proposed by Cruz (2002) may be used to convert industrial size to biological length. The output parameters per carapace length range (CL, mm)

were: number of individuals per carapace length range (N_{CL}), number of total individuals (N_t), mean carapace length, mm (CL), mean weight (Wt), catch in kg (C), and number of undersized lobsters (Nun). Nun was calculated for the years 2009, 2010 and 2018 and compared. In 2018, a processing plant in Fortaleza, Brazil, tested the method above on a simple random sample of lobsters. Lobster samplings were also conducted with the same purpose on fishing boats in GARS in 2001, 2002, 2003 and 2015 (60-100 m depth).

Brazilian government agencies kept yearly updated databases of total lobster landings from 1965 to 2009. A time series of landings per species (red and green) is available for the period 1965-1990 (Ivo, 1996). To estimate the volume of red and green lobster landed between 1991 and 2019, we used an empirical model developed by Cruz et al. (2013a). Currently, data on lobster landings are provided by major fishing companies, exporters and producers affiliated with Sindfrio (a union of fishing industries and cold chains in the State of Ceará), which represented 63% of the national lobster industry in 2018-2020. Figures from non-member fishing companies were obtained from the Comex Stat database (<http://comexstat.mdic.gov.br/es/home>).

The total lobster landings in the Western Caribbean were based on sources compiled by Cruz et al. (1995) for the Bahamas, Nicaragua, Honduras, the Cuban Archipelago, Florida, Caribbean Mexico, Belize, and the Greater and Lesser Antilles. Other sources on landings include (i) the lobster workshop held in Belize (1997), (ii) FAOSTAT (1998-2017), (iii) Appeldoorn & Meyers (1993) for Puerto Rico (1970-1989), (iv) FAOSTAT for Caribbean Colombia (1960-1973), (v) Cruz et al. (2007a), and (vi) Borda & Barreto (2020) for Antillana Enterprise Corporation, Colombia.

Matrix of number of lobsters according to age

Length composition per period (1965-1969, 1970-1979, 1980-1989, 1990-1999 and 2000-2014) was determined by sampling the total catch of the lobster fleet operating on the Brazilian continental shelf. The total landings between 1965 and 2019 take into account the database mentioned above.

We adopted the total weight/carapace length ratio (Wt/CL) established by Cruz et al. (2014): $Wt = 0.0013 * CL^{2.9159}$. The matrix of the number of landed lobsters was obtained by calculating the average weight of the individuals in each length class, then averaging these weights according to the frequency in each length class associated with the corresponding weight of the individuals in that length class. Applying the age-length ratio to each period, we determined the age distribution of the catch. The number of 2-year-old lobsters represented the index of annual recruitment to the fishing grounds. The age structure was estimated with

the growth parameters proposed by Ivo (1996): $L_{\infty} = 231$ mm (CL), $K = 0.235$ (year^{-1}).

The Monte Carlo model (CMSY)

The CMSY model of Froese et al. (2017) was fitted with Brazilian lobster landing data (1965-2019) and resilience values. To determine prior r-ranges (resilience ranges) for the species under assessment, we used the proxies for resilience of the species provided in www.SeaLifeBase.org for invertebrates. Froese & Pauly (2015) categorized resilience as high (0.6-1.5), medium (0.2-0.8), low (0.05-0.5), or very low (0.015-0.1).

Froese's model has been evaluated against 128 stocks in other regions, yielding reasonable predictions of relative biomass and exploitation rates. The equation of the CMSY model is basically:

$$B_{t+1} = B_t + r B_t (1 - B_t/k) - C_T,$$

where B is biomass, k is carrying capacity of the ecosystem, and C_T is catch.

To account for depensation or reduction in recruitment in severely depleted stocks:

$$B_{t+1} = B_t + (4B_t/k)r(1 - B_t/K)B_t - C_T | B_t/K < 0.25,$$

where $(4 B_t/k)$ assumes a linear decrease in recruitment below half the biomass required to attain the maximum sustainable yield (MSY).

First, the unexploited stock size k is larger than the largest catch in the time series. Second, the maximum sustainable catch expressed as a fraction of the available biomass (F_{msy}) depends on the productivity of the stock. Third, the maximum catch will constitute a larger fraction of k in a heavily depleted stock than in a mildly depleted stock. The CMSY model estimates biomass, exploitation rate, MSY and related fisheries reference points based on catch data and resilience ranges.

STOCK ASSESSMENT

Trends in landings and overfishing in Brazil

Brazilian red spiny lobster stocks have been heavily exploited since the 1960s, but landings have fluctuated greatly over time (fig. 2). When landings surpassed the upper limit (6480 t) of the 95% confidence interval of MSY (5570 t), production declined abruptly between 27% and 46% (fig. 2). Large annual fluctuations in recruitment (2-year-old lobsters) ensued, with a tendency to decrease in years of

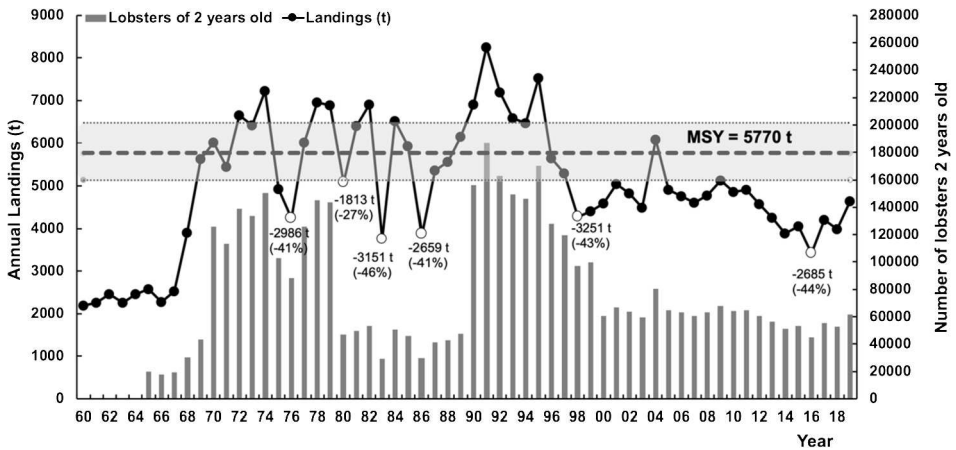


Fig. 2. Fluctuations of annual red lobster landings and number of 2-year-old lobsters. Periods of overfishing (white dots) and reductions in landings (tons and percentage) (negative difference between two consecutive values). The dotted horizontal line represents the maximum sustainable yield ($MSY = rk/4$). The shaded area is the 95% confidence interval (5140-6480 t).

growth overfishing (1976, 1980, 1983, 1986 and 1998). Low levels of recruitment probably led to landings below the lower limit of the 95% CI of MSY (5140 t).

The high proportion of 2-year old lobster in the landings in the 1970s and 1990s may indicate good levels of recruitment, and the biomass trajectory from 1960-2015 seems suitable (fig. 3A). On the other hand, the reduction in landings from 2010 to 2016 results from catch restrictions enforced through management; thus, an increase in biomass would be expected.

The prediction of the relative stock size (B/B_{msy}) shows a narrow range of uncertainty and most of the area is within the 95% confidence interval, with biomass levels high enough to attain MSY. The last two years are very close to the point of equilibrium, indicating that stocks show a tendency for recovery (fig. 3A).

On the other hand, in years with no closed season in lobster fishing (2000-2006), the effort exerted in the 6-month period between November and April represented 40-49% of the annual total. It would therefore seem that the average fishing effort in 2012-2019 was reduced by approx. 45% during the 6-month closed season, corresponding roughly to a 30% decrease in F/F_{msy} .

Paiva (1997) and Fonteles-Filho (1994) suggest that greater efforts result in greater catches. Indeed, a significant association was observed between annual landings and fishing effort (number of traps checked) between 1974 and 1993 ($n = 10$, $R^2 = 0.7035$, $p < 0.001$), and effort was one of the factors responsible for the observed fluctuations in catches. Retrospective analyses reveal a high exploitation rate ($F/F_{msy} > 1$) and no discrepancies in annual predictions (fig.

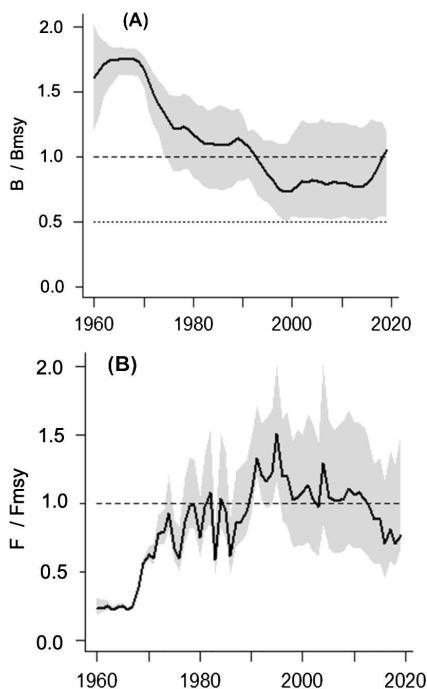


Fig. 3. A, Relative stock size (B/B_{msy}); and B, Relative fishing pressure (F/F_{msy}) per year.

3B). Currently, although the exploitation rate has decreased substantially (0.767), it is still far from the recommended level of 0.5 (Gulland, 1983).

The results obtained with the CMSY model were arranged in a Kobe phase diagram (fig. 4A-D). The area (A) indicates healthy stock sizes (1988-1992) about to collapse from overfishing. The area (B) indicates overfished, with biomass levels too low to produce maximum sustainable yields (1993-2012). The area (C) indicates reduced fishing pressure on stocks and shows a tendency for recovery, despite very low biomass levels, with a probability of 29% (2013-2018).

The relationship between exploitation rate (F/F_{msy}) and stock size (B/B_{msy}) shows a low but sustainable fishing pressure in 2019 (fig. 4D), as a result of the enforced management (fig. 5). The stock is large enough to produce yields close to MSY, with a probability of 58.1%.

The red spiny lobster stocks show a tendency for recovery associated with better monitoring of minimum landing size (≤ 75 mm, CL). In 2009 ($N_t = 730\,262$) and 2010 ($N_t = 294\,374$), the percentage of undersized red lobsters (legal minimum size: 75 mm, CL) was 27.4% and 29.7%, respectively, but in 2018 ($N_t = 3\,810\,117$) it dropped to near zero (0.70%). Also, a lobster sampling conducted at a processing plant in 2018 ($N_t = 862$) yielded a relatively low percentage (3.7%). The values for 2018 show an evident decrease in the percentage of undersized red lobsters.

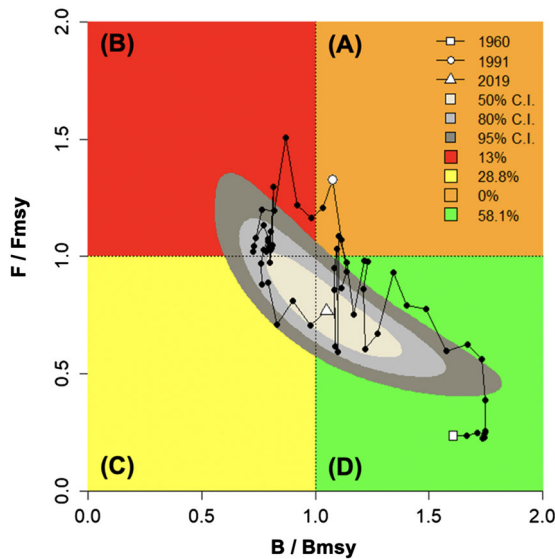


Fig. 4. Kobe diagram: A, orange area: healthy stock sizes threatened by overfishing; B, red area: ongoing overfishing, with insufficient biomass to attain maximum sustainable yields; C, yellow area: reduced fishing pressure on stocks recovering from low biomass levels; D, green area: target area for management, indicating sustainable fishing pressure and a healthy stock size capable of attaining yields close to MSY.

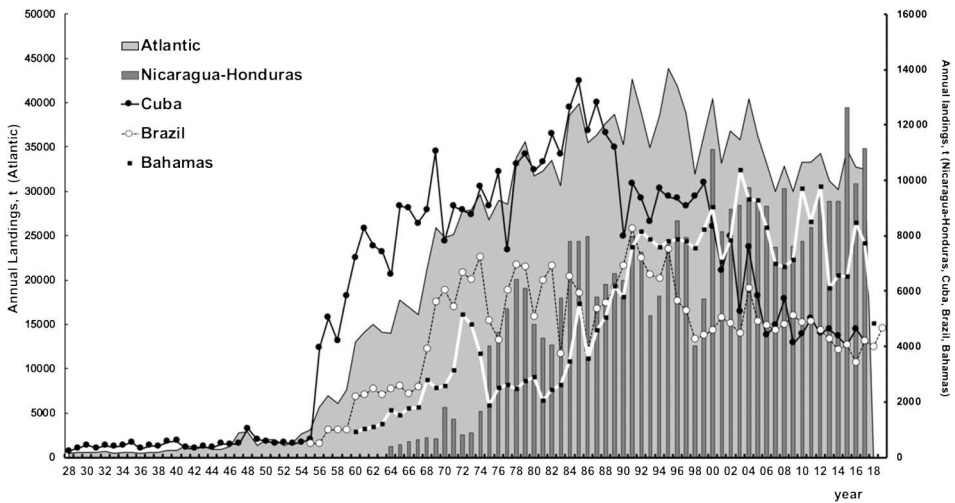


Fig. 5. Annual landings of red spiny lobster, *Panulirus argus* (Latreille, 1804) in the Atlantic by major lobster producers.

In GARS, lobster sampling on fishing boats (60-100 m depth) between 2001 and 2003 and in 2015 revealed fairly low percentages as well: 0.9% ($N_t = 3646$), 4.3% ($N_t = 2066$), 5.1% ($N_t = 1313$) and 6.2% ($N_t = 868$).

However, artisanal fisheries in shallow waters (depth < 15 m) and sales on the local black market, where undersized lobsters make up about 15-20%, are still not adequately monitored, though some improvement is expected.

Large numbers of undersized spiny lobsters have been caught in shallow waters along the Brazilian coast as a consequence of excessive fishing effort (Fonteles-Filho, 2000), leading to severe growth overfishing (Cruz et al., 2013). Currently, there is no evidence of recruitment overfishing of red lobster populations as a result of population reduction (Cruz et al., 2014).

Thus, our findings reveal a scenario of medium resilience (0.559; 95% CI 0.454-0.648), i.e., the ability of the lobster population to withstand sustained adversity, as in the past 20 years of overfishing. We hypothesize that inshore stocks remained resilient to exploitation due to larval subsidies from offshore stocks at greater depths (GARS), with high reproductive potential (Cruz et al., 2014).

Spiny lobster fisheries in the Wider Caribbean-Brazil

Spiny lobster fisheries in the Wider Caribbean and Brazil have gone through four stages: undeveloped (1928-1952), developing (1953-1983), mature (1984-2005), and senescent (since 2006). In 1995, landings in the Atlantic reached a peak of 43 824 t. During the period 1985-2004, commercial landings accounted for 37 900 t on the average, followed by a 20% decrease from 2004 to 2017 due to an abrupt decline in Cuban and Brazilian fisheries since 1990, and in Bahamian fisheries since 2012. From 2013 to 2017, the average production was 32 279 t (fig. 5).

Cuban stocks have been heavily exploited since the 1980s, with a negative tendency despite the apparent recovery in the 1990s. Current patterns of overfishing suggest a poor prognosis (fig. 5). This is due to increased efforts involving highly efficient fishing gear, such as artificial shelters (“pesqueros” or “casitas”) (1309 kg/km², 1988-1992) deployed mostly at the beginning of the fishing season, and traps (“jaulones”) (23 721 kg/km², 1980-1993) deployed during the mass migration from October to February (Cruz & Borda, 2013), compromising catches in the following season (June-September) (Cruz & Adriano, 2001). In the period 1970-1979, annual catches from “casitas” fluctuated around 3488 t, but the figure was much higher (6450 t) in the period 1978-1985, possibly due to an increase in the number of shelters deployed (Cruz & Phillips, 2000). Other likely causes include dramatic environmental fluctuations and impacts, such as from hurricanes (Cruz et al., 2001), and degradation of the coastal marine ecosystem, resulting in reduced carrying capacity (Puga et al., 2018).

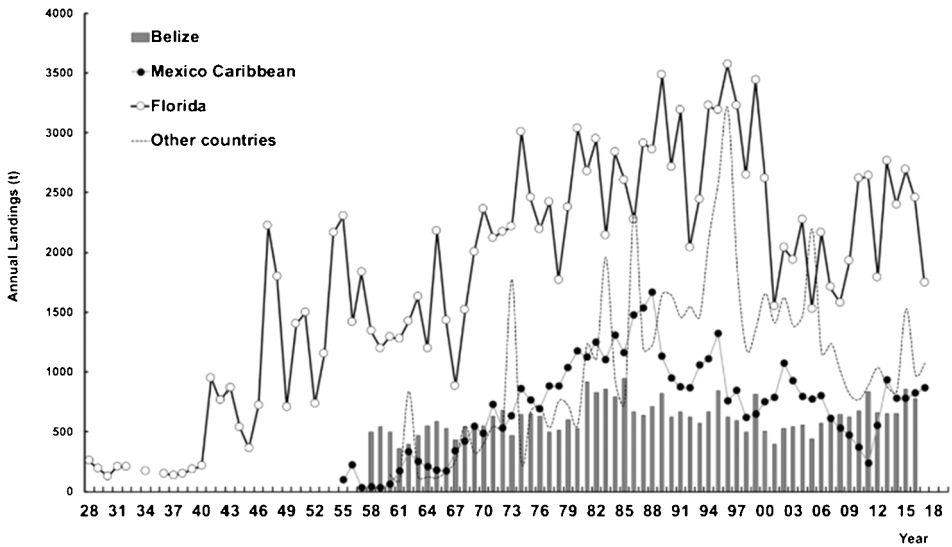


Fig. 6. Annual landings of red spiny lobster, *Panulirus argus* (Latreille, 1804) in Belize, Caribbean Mexico, Florida and other lobster-producing regions.

As a consequence, the reproductive potential of lobsters in the size class 70–80 mm CL decreased significantly (1998–1996) (Cruz et al., 2014) and migration patterns and recruitment were severely compromised, with landings falling to pre-1950 levels. This is the consequence of the lack of management on part of decision makers who ignored the recommendations of the scientific community based on lobster studies.

Lobster stocks off the Bahamas are heavily exploited or overfished, but surveys conducted by multiple researchers have so far yielded inconsistent data (Gascoigne et al., 2018). In Nicaragua, assessments have shown a decline in mean length — a common indication of overfishing (Valle-Esquivel, 2011). Very little is known about the situation of the lobster stocks off Honduras (FAO, 2017).

Florida, the Mexican Caribbean, Belize and other Caribbean countries report landings from 500 to 4000 t (fig. 6). Florida's stock is at risk of collapsing (Buesa, 2018). At an average of 2409 t, Florida is the largest producer in the region, accounting for 7.46% of total landings (2013–2017), followed by the Mexican Caribbean (836 t; 2.59%) and Belize (733 t; 2.27%) in the same period. The remainder of the Caribbean countries total 1056 t on the average. All these countries experience high annual variations in landings, possibly associated with fluctuations in recruitment and fishing effort.

Based on annual average, in the period 2013–2017 the top five spiny lobster producers were the Bahamas (7074 t), Brazil (5533 t, red + green lobsters), Nicaragua (5215 t), Honduras (5207 t), and Cuba (4362 t).

Fishing effort and stock assessment

Spiny lobster fishing has historically been carried out with traps, gill nets and scuba diving on artificial shelters. Eventually, the uncontrolled rise in fishing effort led to an abrupt drop in production. However, the absence of conversion factors between types of fishing gear makes it impossible to arrive at a standard unit of effort. Between the 1960s and the 1990s, the number of checked traps was used for this purpose, but variables related to fishing fleet dynamics and cost should also be considered when assessing catch and effort (Fonteles-Filho, 1994).

Since the 1990s, lobster fishing statistics have been patchy and inconsistent, compromising all attempts at analysing commercial stocks and trends in the fishing sector. In the last 20 years, no information on effective effort has been available for Brazilian commercial lobster fisheries, and stock assessment has been based on the parameter “fishing days”, which fails to provide a reliable estimate of abundance (N) and effort (Sparre & Venema, 1995). Such assessments tend to underestimate fishing mortality and do not take into account the increase in efficiency that has occurred over the years.

The growing use of scuba diving to catch spiny lobsters on artificial shelters (“marambaia”) in shallow waters along the Brazilian coast has increased the number of undersized individuals captured (>50%) (Fonteles-Filho, 2000), leading to severe growth overfishing (Cruz et al., 2013b). On the other hand, there is the intensive exploitation of spiny lobster stocks in GARS (50-100 m), 50% or more of which consists of mature adults with high reproductive potential (Cruz et al., 2014). Along with the capture of berried females for more than 40 years, this situation created a negative trend in lobster landings between 2004 and 2016. Currently, there is no evidence of recruitment overfishing of the red lobster stock.

Over the last decade, the spiny lobster stocks on the Brazilian continental shelf have been assessed with several models. In the general production yield model, using catch (t) per million traps-day as CPUE, the MSY of red lobster was estimated to be 6862 t (Fonteles-Filho, 2000) and 6993 t (Fonteles-Filho, 2007). Cruz et al. (2013) proposed a new approach for the assessment of wild spiny lobster stocks using a fishing intensity index (checked traps per km² per year) and a productivity index (kg of landings per km² per year). The method produced a rough estimate of 9381 t potential yield for the period 1974-1991. The Bayesian stock assessment model (Andrade, 2015) also uses catch (t) per million traps-day as CPUE. Based on datasets from the period 1965-1997 and isolated data from 2002 and 2004, the model shows that stocks have been heavily overexploited since the 1980s and may be close to a collapse. Sant’Ana & Kinas (2019) used a biomass dynamics model with a Bayesian approach, involving two CPUE time series (kg/trap × day, and kg/day of gill net) and three scenarios. The

authors concluded that the stock is overfished, having reached a level of biomass insufficient for maintaining the stock. Recent investigations (Aragão & Cintra, 2019) of sequential analysis of fisheries, covering only eleven years (2005 to 2015), concluded that, despite high fishing mortality rates and a strongly depleted spawning biomass, the stock still retains some stability. The authors calculated the nominal effort (f) in terms of number of days at sea ($f = \text{national catch (kg)}/\text{CPUE (gillnets)}$)).

As shown by the assessments above, there is much uncertainty about the current state of the stock. Differences in scientific methodology, criteria and parameters compromise the comparability of the available studies and datasets. For example, several researchers have argued that CPUE is not a valid index of relative abundance. Further research is required to clarify the dynamics of fishing gears and the fluctuations in lobster abundance in the stock. In any case, the present-day scenario is promising, with a tendency for recovery of red lobster stocks.

Information on catch and effort by gear is at present inconsistent and must be improved to make it useful in stock assessment models (Cruz et al., 2020). Cruz & Borda (2013) concluded that CPUE was not proportional to abundance (hyperstability) in Cuban traps and in natural shelters exploited by divers in Caribbean Colombia. A similar lack of proportionality between CPUE and N was observed in a preliminary study from Northeastern Brazil (Barroso, 2012) on red lobsters caught with gill nets (“*caçoeira*”) and traps (“*manzuá*” and “*cangalha*”). This is a common form of non-proportionality, according to Hilborn & Walters (1992).

Harley et al. (2001) discussed the inappropriateness of using CPUE data based on strong evidence of hyperstability in three groups of fish, and many other authors have pointed out problems with this index (Beverton & Holt, 1957; Paloheimo & Dickie, 1964; Gulland, 1974; Hilborn & Walters, 1992; Harley et al., 2001; Walters et al., 2003; Maunder et al., 2006).

NATURAL MORTALITY

Lobster predation, diseases and natural mortality

The study of population dynamics involves a broad set of parameters, one of which is natural mortality (M). M is very difficult to estimate as it depends on current (and often very limited) knowledge of disease, predation, cannibalism, environmental fluctuations, etc. Since the 1950s, empirical studies have tested equations to determine M (cf. Laurec & Le Guen, 1981; Sparre & Venema, 1998; Cadima, 2000), reporting a large annual variability from 0.20 (Waugh, 1980) to 1.03 (Munro, 1974) for exploited populations.

Pauly (1980, 1983) developed two empirical equations to calculate M for schooling pelagic or demersal fish. Based on Pauly's M equation for fish, Cruz et al. (1981) proposed an equation specifically for spiny lobsters (Palinuridae and Nephropidae): $M = -0.0277 - 0.0004 * L_{\infty} + 0.5397 * K + 0.0119 * T$ (L_{∞} = mm CL; K = annual growth; T = average annual temperature at the surface in °C). Fitting the growth parameters of Ivo (1996) and González-Cano & Rocha (1995) into this equation yields an M rate for Brazilian lobsters of, respectively, 0.35 year⁻¹ and 0.36 year⁻¹, matching the M rate of 0.36 year⁻¹ reported by Fonteles-Filho (2000). The latter author calculated M for green lobsters (0.39 year⁻¹) in separate. In comparison, the M rate for mature lobsters in the Cuban archipelago was between 0.30 and 0.35 year⁻¹, although Buesa (1972) reported a lower value (0.26 year⁻¹) for this area based on mark-recapture data.

In order to evaluate the M rate for juveniles in nursery areas, we analysed the occurrence of illnesses and potential predators. The behaviour of adult fish (groupers, puffers, snappers, triggerfish) occasionally inhabiting concrete block shelters for juvenile lobsters was studied during the day, but the fish were not observed to prey on lobsters. We then concentrated our observations (360 days, 1982-1984) on nurse sharks (*Ginglymostoma cirratum* Bonnaterre, 1788). The sharks spent 90% of the daytime inside the shelters, and when the stomach contents were examined, 62% ($n = 21$) had a full stomach containing on average 50% juvenile lobsters, 25% small fish, 18% small shrimp and crabs, and 7% small octopus. The Index of juvenile predation was 0.0032, leading to a juvenile M rate of 0.23 year⁻¹ (cf. Cruz et al., 2006).

Cruz et al. (2006) sampled 11 722 juveniles in artificial shelters in a nursery area to quantify disease occurrence. The literature on carapace disease in *P. argus* is scarce and no previous studies on the microbial flora of the exoskeletal structure of these organisms were found by Porter et al. (2001) or Porter (2004).

The term 'shell disease' refers to a series of necrotic, erosive lesions, excavations and/or discolorations in the cuticle of the exoskeleton of spiny lobsters (Fisher et al., 1978; Cobb & Castro, 2006; Feinman et al., 2017). Porter (2004) cites the appearance of lesions indicative of shell disease, both in captivity associated with poor water quality and agglomeration (prior to 1998) and in the natural environment, as reported by fishermen. In a survey carried out that same year, 19% of the lobsters caught in traps displayed signs of shell disease. Unlike Porter (2004), we have never observed shell disease in juvenile lobsters (Cruz et al., 2006).

Gaffkaemia is a potentially deadly lobster disease caused by the bacterial species *Aerococcus viridans* var. *homari*, a variety of *Aerococcus viridans* Williams, Hirsch & Cowan, 1953. Infected individuals become lethargic, the abdomen acquires a pinkish hue and the haemolymph turns dense and milky. When captured, affected lobsters die after a few minutes on deck (Cruz et al., 2006). The

cuticle serves as a natural barrier against the pathogen; thus, damage to the animal's exoskeleton appears to be a precondition for infection (Fisher et al., 1978; Stewart, 1984; Shields, 2011).

Bobes et al. (1988) isolated and identified *A. viridans* var. *homari* in six specimens of *P. argus* with symptoms of shell disease. Cruz et al. (2006) found juveniles to be extremely susceptible to gaffkaemia. Infected juveniles had a carapace length of 20-40 mm and the index of infected juveniles was 0.0027, leading to a juvenile M rate of 0.19 year^{-1} . No infection was seen in the puerulus (4-6 mm CL) or algal stage (6-16 mm CL).

The total natural mortality rate of juveniles in the nursery areas off Cuba has been estimated at 0.42 year^{-1} , with predation ($M = 0.23 \text{ year}^{-1}$) and disease ($M = 0.19 \text{ year}^{-1}$) as the major causes of death in *P. argus* (cf. Cruz et al., 2006). This is compatible with the rate of 0.44 year^{-1} calculated by Cruz et al. (1986b) during a mark-recapture programme for juveniles.

Prior to the observations of Cruz et al. (2006), estimations of juvenile M rates did not take into account causes of death, such as disease and predation (spiny lobsters are not known to engage in cannibalism). The above study was carried out in Cuba but may serve as a reference for Brazilian studies on natural mortality. The results of the study point to a density-dependent mortality, with different M rates for juveniles and mature lobsters and decreasing rates with increasing age.

EXTERNALITIES AND MARKETING IN LOBSTER FISHERIES

Externalities associated with lobster fisheries

The rebuilding of lobster stocks and fisheries in NE has produced various outcomes related to fishermen, fishing grounds, vessels, routines and methods, in addition to trends in landings and markets (cf. Santana et al., 2015). The new knowledge on fishing externalities, that is defined as any external effect not accounted for by the fisherman who generates it, but which does affect other users of the resource (Seijo et al., 1997), can be of an environmental, fishery, social, market or economic nature. The market inefficiencies and management options have helped prevent negative impacts on lobster resources associated with the life cycle. So far, it is the most successful attempt at recovering and rebuilding Brazilian lobster stocks.

The network of lobster production and wholesale marketing is quite complex: vessels of varying size and different legal or illegal methods and fishing gear are used, none of which are selective. Also, lobster fishermen rely on middlemen to sell their products. These market inefficiencies are economically detrimental to fishermen, whether self-employed or hired, and have mostly turned lobster fishing

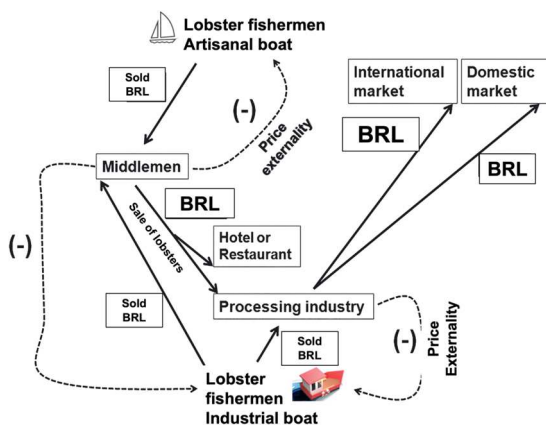


Fig. 7. Lobster supply chain. Lobster fishermen operating in shallow and deep waters, respectively, are affected by negative price externalities generated by middlemen and the processing industry. Differences in sales prices are indicated by differences in the size of the symbol BRL. [Modified from Santana et al., 2015.].

into a subsistence activity (fig. 7). The acceptance of meager profits on the part of artisanal fishermen is a classic example of what psychologists call “the social trap” (Platt, 1973): a situation in which a cooperative solution with obvious social benefits cannot be reached due to lack of confidence, education and financial maturity.

In general, Brazilian fishermen disregard regulations on resource use and access. This is equivalent to a regime of non-property (open access) in which individuals can directly take possession of resources. Unrestricted access leads to overexploitation of common property, making rational allocation of resources impossible and generating negative externalities.

The average size and range of lobsters by depth and location reveal a stratified pattern. In GARS, large and mature individuals are more abundant, with the greatest percentage of berried females (57%; $n = 2452$) throughout the year (fig. 8). A similar positive association between depth and the size of sexually active females was observed in Bimini (Kanciruk & Herrnkind, 1976), Florida (Bertelsen, 1999) and Cuba (Cruz & Phillips, 2000). As reported by Silva et al. (2013), 104 specimens of slipper lobsters (*Parribacus antarcticus* Lund, 1793; *Scyllarides delfosi* Holthuis, 1960) with an average size of 90.16 mm CL (range: 30-100 mm) were caught incidentally during commercial red lobster fisheries in GARS. During another prospection, the authors captured other lobster species at depths between 41 and 626 m.

The fishing methods and gear described above are associated with several types of externalities. The artisanal fishing fleet (fig. 8A) exploits primarily juveniles and pre-recruits in shallow waters (3-5 m) at the beginning of the open season, and

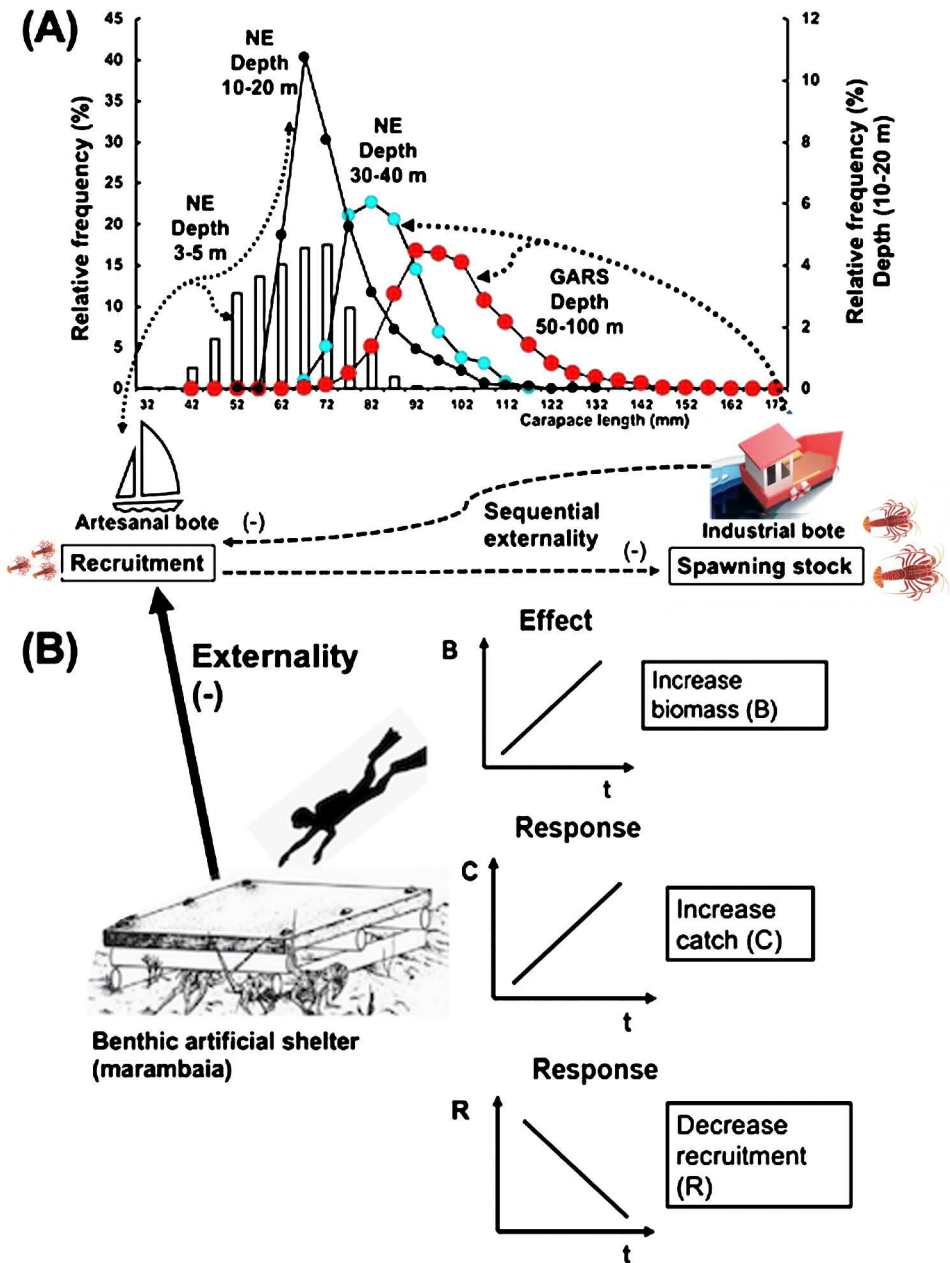


Fig. 8. Summary of the dynamics of lobster fisheries on the Brazilian continental shelf. A, Impact of the artisanal fleet and the industrial fleet on the size structure of the lobster stock; B, impact of artificial shelters (“marambaia”) on stock structures, thereby generating a sequential negative externality, that occurs when artisanal and industrial fleets exploit different components of the lobster stock. [Modified from Santana et al., 2015.].

later at 10-20 m. In a catch examined by Cruz et al. (2013b), the distribution was 60% young animals (50-73 mm CL), 16% pre-recruits (74-79 mm CL), 9% mid-sized adults (80-99 mm CL) and 3% large adults (>100 mm CL). The industrial fishing fleet exploits adult lobster stocks at intermediate depths (30-40 m) using traps/gill nets, and in deeper waters (GARS) using mostly gillnets. The lack of control of the fishing fleet has negative repercussions on the size structure of the lobster stock, compromising recruitment and spawning and generating mutually negative externalities for artisanal and industrial producers (fig. 8A).

Artificial shelters also generate environmental externalities, some of which are negative (e.g., disturbances in the benthic ecosystem, damaging the biota under the shelter) (Cruz et al., 1986a; Cruz & Phillips, 2000), but most environmental externalities are positive: increased feeding efficiency and growth rates, increased availability of shelter (Cruz & Borda, 2013) and reduced predator pressure (Eggleston et al., 1990) (fig. 8B). Over time, biomass and productivity tend to increase while the cost and effort of lobster fishing decrease. Although the capture of juveniles affects recruitment and stocks negatively, the use of lobster shelters may be recommended under certain environmental and socioeconomic conditions and if managed and regulated with proper enforcement (Cruz & Phillips, 2000; Conrad & Danoff-Burg, 2011; Cruz & Borda, 2013).

Fishing, marketing and exports of whole lobster

Efforts should be directed at increasing the production of whole lobster without causing mechanical damage, provided the appropriate know-how is transferred to the communities engaged in the production chain.

The markets targeted by producers in the Wider Caribbean and Brazil include the U.S., France, Spain, Italy and China (Hong Kong). The main exporters (with many markets in common) are the Bahamas, Nicaragua, Honduras, Cuba and Brazil. Average annual exports in 2009-2014 amounted to USD 249 million, 71% (USD 181 million) of which to the U.S. and 29% (USD 68 million) to Europe (France, Spain). Cuba is one of the few Caribbean countries exporting to Hong Kong (currently at USD >30/kg). The price of this highly valued commodity varies greatly from one country to another; e.g., USD 90/kg in New Zealand and USD 40/kg in South Africa (FAO, 2017).

The Bahamas and Brazil are the main exporters to the U.S., followed by Honduras and Nicaragua. Brazilian exports of whole lobster have been on the rise since 2013, and in 2019 totalled USD 91.8 million (fig. 9A), of which the northeastern State of Ceará accounted for USD 58.3 million. In 2019, Brazil exported frozen lobster tails to 21 countries, especially to the U.S. (86%), Australia (9%), Japan (2.2%), New Zealand (1.9%) and China (1.1%) (the remaining 16

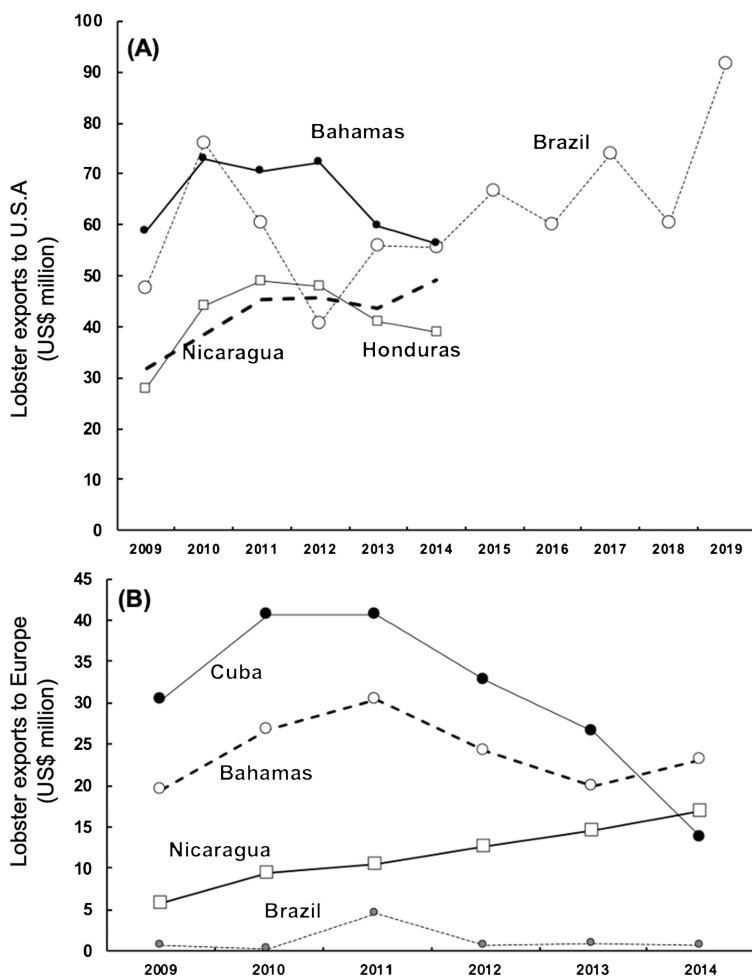


Fig. 9. Annual lobster exports to: A, the United States; and B, Europe (France and Spain), by major exporting countries.

countries make up 1%) (Comex Stat, 2019), but Brazilian sales management in Europe and China is still not competitive. The lobster industries of the Bahamas, Nicaragua and Honduras recently made a joint effort to gain access to the European market, while Cuba reduced exports as a consequence of overfishing and Brazil remains in an unfavourable position (fig. 9B). The Brazilian lobster industry needs to rethink its strategy in order to become competitive in the European and Asian markets.

To establish the new paradigm of whole lobster exports, the Brazilian production chain needs urgent transformation, with innovations in management. The lobster fleet will have to adopt new methods of capture and onboard handling, storage and transportation, and middlemen and suppliers will need to learn how to keep

the catch viable until the time of sale or delivery at the processing plant. In addition, a project sponsored by the Federal Institute of Education, Science and Technology of Ceará (Santana et al., 2019) proposes to replace inefficient lobster fishing vessels with catamarans, which allow to improve performance, humanize fishing operations, store live lobsters and, consequently, raise fishermen's living standard. Finally, training courses in fishing techniques, vessel operation, handling, storage and licensing have been administered to the younger generations of the local fishing communities.

MANAGEMENT AND LIFE CYCLE

Implications of the lobster life cycle on management

Resources need to be managed much more efficiently than they have been in the past. The closed season (November to April) currently protects spiny lobsters during sensitive stages of their life cycle: spawning, recruitment to nursery areas and recruitment to fishing grounds at an average size of 76.5 mm CL (range: 74-79) (fig. 10). These results are very consistent. No additional fieldwork on the directed offshore movement of juveniles (*Panulirus argus*) has been published and this recruitment size offers a new perspective on the regulation of pre-recruit lobsters.

Landings reach a peak at the start of the open season in May, followed by a decrease in production. Approximately 40% of the total annual lobster catches occur in May and June (fig. 10E) (Cruz et al., 2011).

In the beginning of the open season, the artisanal lobster fleet operates in shallow waters where pre-recruits represent between 15% (May) and 18% (June) of the population. During this period, intense supervision is required to prevent fishing with illegal methods and to enforce the law against violators. Considering the life cycle of the spiny lobster, the closed season should be changed from November-April to January-June, since May and June is when pre-recruits are most abundant.

Lobster fishing and marketing management measures are currently being implemented. Thus, in the first week of the closed season, only transportation, storage and sale on the domestic market will be allowed. At the start of the open season, the catch delivered to processing plants by fishermen or middlemen must consist of at least 95% whole lobsters (the remainder being tails). These measures will help prevent the exploitation of undersized lobsters (tails) and berried females and inhibit the black market and fishing during the closed season. In addition, we recommend implementing the following management policies:

- Lobster traps should be furnished with escape gaps for undersized lobsters.
- During the open season, fishing quotas of 4000 t for red lobsters and 900 t for green lobsters should be enforced (although the setting of quotas on catches is in itself insufficient to protect lobster stocks).

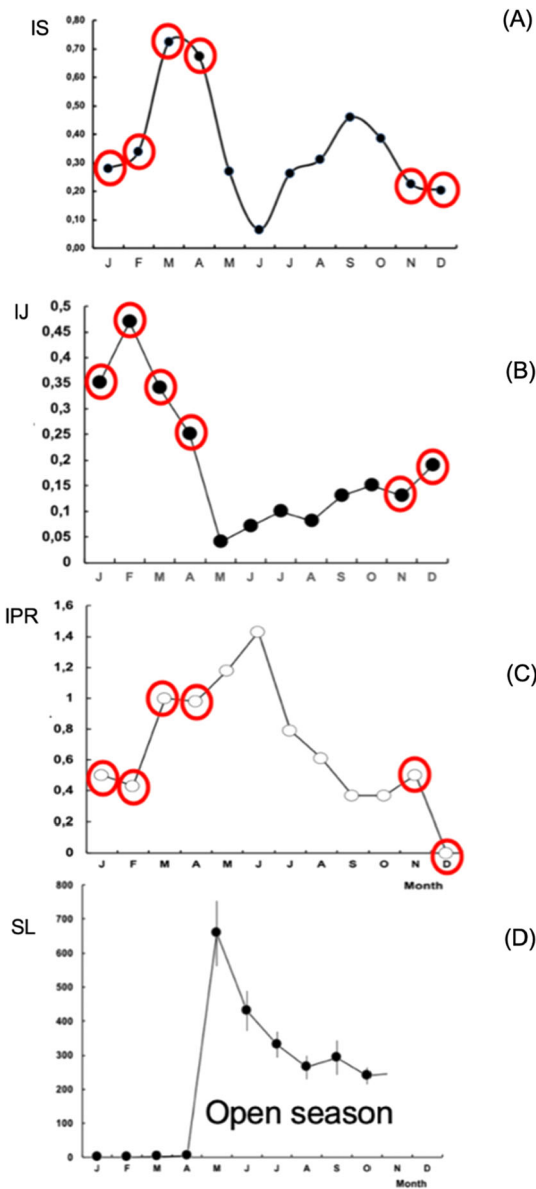


Fig. 10. Theoretical spiny red lobster recruitment pattern (cf. *Panulirus argus* (Latreille, 1804)) on the continental shelf of Brazil, showing the development of a cohort from birth, represented by the reproduction cycle: A, spawning season; B, recruitment to the nursery area; C, recruitment to the fishing ground (fishery recruitment); and D, seasonal landings. Red circle: closed season (November-April); black dots: open season (May-October). IS, index of spawning; IJ, index of juveniles; IPR, index of pre-recruits; SL, seasonal landings (t).

- Considering the longevity and absence of reproductive senility in lobsters, a maximum catch size of 135 mm (CL) for red lobsters and 100 mm (CL) for green lobsters should be enforced. This would protect the stock's reproductive potential and avoid disturbing sensitive habitats (Cruz et al., 2014).

CONCLUSION OF MAJOR FINDINGS

The enforcement since 2012 of a 6-month closed season (November-April) has decreased the fishing effort by approximately 45%, favouring reproduction, juvenile recruitment and the migration of pre-recruits to fishing grounds. The prediction of the relative stock size (B/B_{msy}) shows a narrow range of uncertainty and most of the area is within the 95% confidence interval, with biomass levels high enough to attain MSY. The last two years are very close to the point of equilibrium, indicating a tendency for recovery. In 2019, the stock size was capable of producing yields close to the maximum sustainable yield (5770 t), with a probability of 58.1%. The exploitation rate has decreased considerably (0.767) but is still far from the recommended level of 0.5.

The Brazilian lobster industry needs to change its strategy to become competitive on the whole lobster market in Europe and Asia. To do so, lobster fishermen, suppliers and processing industries need to adopt new methods of capture, handling, storage and transportation, among other things by replacing inefficient lobster fishing vessels with catamarans, as proposed by local projects.

Assessments of fishing resources based on incomplete or distorted artisanal fishing data are misleading. But recruitment and potential yield can be predicted using indices independent of fishing data based on knowledge of the life cycle.

Efforts should be focused on developing and implementing truly efficient management measures. Current regulations are more or less aligned with our knowledge of the life cycle of Brazilian red lobster populations. Field studies on the lobster life cycle have widened our understanding of recruitment patterns in NE and GARS.

The enforcement of management measures should be continued in order to eliminate sources of negative externalities and rebuild stocks. Some of the most urgent actions include (i) eradication of gillnets ("caçoeira") and artificial shelters ("marambaia"), (ii) prohibition of capture of berried females and undersized individuals (red lobster <75 mm CL; green lobster <65 mm CL), (iii) limitation of fishing methods to traps; (iv) permanent fishing ban on juvenile nursery areas; (v) prohibition of use of undersized lobsters as decoys in traps; and (vi) resumption of professional collection of lobster fishing data to subsidize future analyses of commercial stocks and trends in the sector.

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